

FitForge

Complete System Manual

Your move, your progress.

FitForge is an all-in-one gym management and workout tracking platform. It combines workout logging, routine building, analytics, membership management, equipment rentals, and personal coaching into a single web application. This manual documents every feature, the technical architecture, and step-by-step setup instructions.

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Table of Contents

1. Overview
2. System Architecture
3. Installation & Setup
4. User Features
5. Coach Features
6. Administrator Features
7. API Reference
8. Database Schema
9. Troubleshooting
10. Appendix

1. Overview

1.1 What is FitForge?

FitForge is a full-stack web application designed for gym members, coaches, and administrators. It tracks workouts, manages gym memberships, handles equipment rentals, and facilitates personal coaching — all in one platform. The application uses a hybrid database architecture: MySQL stores structured, relational data, while MongoDB stores flexible, high-volume event data.

1.2 Target Users

- Gym Members — track workouts, build routines, view progress, subscribe to memberships, rent equipment, receive coaching.
- Coaches — view assigned clients' workout history, analyze progress, and send detailed feedback with ratings.
- Administrators — manage users, verify payments, oversee memberships and rentals, manage inventory, view analytics.

1.3 Key Features

Feature	Description
Workout Tracking	Log sets, reps, and weight. Automatic PR detection using Epley 1RM formula.
Routine Builder	Create reusable workout templates with exercises, sets, and reps.
Analytics Dashboard	Volume charts, muscle balance breakdown, and personal record table.
Exercise Library	42+ exercises filterable by muscle group and equipment.
Membership System	Walk-in, monthly, and yearly plans with expiry tracking.
Payment Processing	GCash, Maya, bank transfer, card, and cash. QR verification via email.
Equipment Rentals	Hourly reservations with live stock tracking and receipts.
Coach System	Coach-client assignments, feedback with ratings, unread badges.
Admin Dashboard	User management, revenue analytics, payment verification.
Hybrid Database	MySQL for relational data, MongoDB for logs and payments.

1.4 Technology Stack

Layer	Technology
Backend	Node.js + Express

Frontend	React 18 + React Router + Recharts
Styling	Tailwind CSS v4
Primary DB	MySQL 8 (via XAMPP)
Secondary DB	MongoDB Atlas (cloud)
Auth	JWT + bcrypt
Email	Nodemailer + Gmail SMTP
QR Codes	qrcode npm library

2. System Architecture

2.1 High-Level Architecture

FitForge follows a three-tier architecture. The React frontend runs in the browser and communicates with the Express backend via REST API calls. The backend connects to two databases: MySQL for structured data and MongoDB for document data. Authentication uses JWT tokens stored in the browser's localStorage.

2.2 Database Split

The hybrid architecture separates concerns based on data characteristics. MySQL handles everything that must be exact, relational, and transactional. MongoDB handles logs, events, and documents whose shape varies per record.

Concern	Database	Reason
Users, roles, passwords	MySQL	Unique constraints, foreign keys
Workouts, routines, sets	MySQL	Heavy joins, transactional inserts
Memberships	MySQL	Money and dates, ACID guarantees
Rentals + inventory	MySQL	Atomic stock decrements
Coach assignments	MySQL	Relational, foreign keys
Activity logs	MongoDB	Append-only, high volume, flexible metadata
Payment receipts	MongoDB	Variable fields per method
Daily metrics	MongoDB	Pre-aggregated rollups

2.3 Authentication Flow

Users register with email and password. Passwords are hashed with bcrypt (10 rounds) before storage. On login, the server verifies the password and issues a JWT containing the user's id, email, and role. The frontend stores this token in localStorage and attaches it to every subsequent request via an Axios interceptor. Protected routes verify the token's signature and expiry.

2.4 Payment Verification Flow

For GCash, Maya, and bank transfers, the payment is created with status 'pending'. A QR code encoding the verification URL is generated and emailed to the user. The user scans the QR on their phone, confirms they've sent the money, and the admin verifies the payment. Cash and card payments skip verification and activate immediately.

3. Installation & Setup

3.1 Prerequisites

- Node.js 18+ and npm
- XAMPP (MySQL) — <https://www.apachefriends.org>
- MongoDB Atlas account (free tier) — <https://cloud.mongodb.com>
- Gmail account with 2FA enabled (for SMTP)
- Git (optional)

3.2 Clone and Install

```
git clone <repository-url> FitForge
cd FitForge
npm install
cd client
npm install
```

3.3 Environment Configuration

Create a .env file in the project root with these variables:

```
# Server
PORT=5000
NODE_ENV=development

# MySQL (XAMPP defaults)
MYSQL_HOST=localhost
MYSQL_PORT=3306
MYSQL_USER=root
MYSQL_PASSWORD=
MYSQL_DATABASE=fitforge

# MongoDB Atlas
MONGO_URI=mongodb+srv://user:pass@cluster.mongodb.net/fitforge

# JWT
JWT_SECRET=<64-char-random-string>
JWT_EXPIRES_IN=7d

# Gmail SMTP
EMAIL_USER=your.email@gmail.com
EMAIL_PASS=<16-char-app-password>
EMAIL_CC_ADMIN=

# Client URL
CLIENT_URL=http://localhost:5173
```

3.4 Gmail App Password Setup

- Go to myaccount.google.com/security

- Enable 2-Step Verification (required)
- Go to myaccount.google.com/apppasswords
- Create a new app password for 'FitForge Mailer'
- Copy the 16-character password into EMAIL_PASS

3.5 Database Setup

Start MySQL in XAMPP Control Panel, then run the SQL schema in phpMyAdmin:

1. Open <http://localhost/phpmyadmin>
2. Click the SQL tab
3. Paste the full schema from Section 8
4. Click Go

Verify with:

```
SHOW TABLES;  
-- Should return 12 tables
```

3.6 Start the Application

Open two terminals:

```
# Terminal 1 - Backend  
npm run dev  
# Runs on http://localhost:5000  
  
# Terminal 2 - Frontend  
cd client  
npm run dev  
# Runs on http://localhost:5173
```

Open <http://localhost:5173> in your browser.

3.7 Create Your First Admin

- ```
Option A: Register via UI then promote
1. Go to http://localhost:5173/register
2. Create an account
3. In phpMyAdmin, run:
 UPDATE users SET role='admin'
 WHERE email='you@example.com';
4. Log out and log back in
```

## 4. User Features

### 4.1 Registration & Login

New users register at /register with email, password, and optional username, full name, and birthday. Passwords must be at least 6 characters. After registration, the user is automatically logged in.

### 4.2 Profile Page

The profile page (/profile) shows the user's name, username, email, birthday, age, role, and membership status. Four summary cards show total spent, refunded amount, transaction count, and active rentals. Four tabs organize content: Overview, Transactions, Memberships, and Rentals.

Active membership appears as a prominent banner with:

- Plan name (Walk-in / Monthly / Yearly)
- Start and end dates
- Days remaining (color-coded: green > 7 days, orange <= 7 days, red if expired)

### 4.3 Logging Workouts

From the Workouts page:

- Click '+ Log workout'
- Enter workout name (e.g., 'Push Day') and date
- Add sets - select exercise, weight, reps
- Click 'Save workout'

The system automatically computes estimated 1RM using Epley's formula:  $\text{weight} \times (1 + \text{reps} / 30)$ . If a new record is set, a toast confirms 'N new PR(s)!'.

### 4.4 Building Routines

From the Routines page:

- Click '+ New routine'
- Name the routine (e.g., 'Leg Day')
- Add exercises with target sets and reps
- Click 'Save routine'

Click 'Start' on any routine to jump to the Workouts page with the form pre-filled from that routine's exercises.

### 4.5 Analytics

The Analytics page shows three sections:

- Volume over time - line chart of total kg lifted per day
- Muscle balance - pie chart of volume per muscle group
- Personal records - table of best 1RM per exercise

Range selector controls the time window (7, 30, 90, 365 days).

## 4.6 Exercise Library

Browse 42+ exercises. Filter by muscle group (Chest, Back, Legs, Shoulders, Arms, Core) or equipment type (Barbell, Dumbbell, Machine, Cable, Bodyweight). Search by name.

## 4.7 Membership

Three plans are available:

| Plan    | Price      | Duration |
|---------|------------|----------|
| Walk-in | PHP 100    | 1 day    |
| Monthly | PHP 1,200  | 30 days  |
| Yearly  | PHP 12,000 | 365 days |

Payment methods: Cash, GCash, Maya, Bank transfer, Card.

GCash, Maya, and Bank payments trigger QR code generation and an email with a verification link. Cash and Card payments activate immediately.

## 4.8 Equipment Rentals

Browse the rental catalog and reserve equipment by the hour:

- Olympic Barbell - PHP 50/hr
- Power Rack - PHP 120/hr
- Dumbbell Set - PHP 80/hr
- Resistance Bands - PHP 25/hr
- Kettlebell (16kg) - PHP 40/hr
- Treadmill - PHP 150/hr
- Rowing Machine - PHP 130/hr
- Cable Machine - PHP 140/hr

Stock is decremented on reservation and restored on return. Each rental generates a receipt showing item, duration, rate, and total cost.

## 4.9 Personal Coaching

If assigned to a coach by an admin, users see a 'My Coach' link in the navbar. This page shows the assigned coach and all feedback they've sent. Feedback includes subject, content, optional star rating,

and the linked workout. Unread items have a blue NEW badge.

## 5. Coach Features

### 5.1 Coach Dashboard

Coaches see a dedicated dashboard at /coach showing all assigned clients as cards. Each card displays the client's name, email, workout count, last workout date, and any unread feedback count.

### 5.2 Client Detail View

Clicking a client card opens their detail page with four sections:

- Client header - name, email, membership status
- Stats grid - workout count, PRs, feedback count, member since
- Workout history - expandable list showing all sets
- Recent PRs - last 5 personal records

### 5.3 Sending Feedback

On the client detail page, the feedback form allows:

- Subject (optional)
- Workout attachment (optional - links feedback to a specific workout)
- Star rating (optional - 1 to 5 stars)
- Content (required - the actual coaching notes)

Feedback appears immediately in the client's 'My Coach' page. Coaches can delete their own feedback but not other coaches'.

## 6. Administrator Features

### 6.1 Admin Dashboard Overview

The admin dashboard has 10 tabs accessible from the top navigation:

### 6.2 Profile Tab

Shows the admin's own account card and a grid of all users who've used the website.

### 6.3 Stats Tab

Three KPI cards (total users, total workouts, logins in range) plus a stacked bar chart of platform activity. Range selector: 7, 30, 90, or 365 days. Below the chart, the Revenue section shows daily revenue line chart, payment method pie chart, and purpose breakdown.

### 6.4 Users Tab

Full user table with ID, name, email, role, and join date. Actions per user:

- View - opens the user detail modal
- Make coach - promotes user to coach role
- Reset to user - demotes coach/admin back to user

### 6.5 Activity Tab

Reverse-chronological feed of all activity log events from MongoDB: logins, registrations, workouts, payments, coach feedback.

### 6.6 Payments Tab

Four KPI cards (total payments, net revenue, refunded amount, unique users) and a table of every payment. Click 'View' to see the receipt.

### 6.7 Pending Tab

Verification queue for GCash, Maya, and Bank payments. Shows user, reference, amount, method, and whether the user has confirmed. Actions:

- Resend - re-sends the QR verification email (can CC additional addresses)
- Verify - marks payment as paid and activates membership if applicable
- Reject - marks payment as failed and notifies user

### 6.8 Transactions Tab

Two tables side by side. Filter by user dropdown at the top.

Memberships table - user, type, period, current status, and a dropdown to change status to:

- Active - reactivates and un-refunds payments
- Cancelled - cancels without refund
- Cancelled + Refund - cancels and marks payment as refunded

Rentals table - user, item, quantity, total, status, and a dropdown to change status to:

- Reserved, Confirmed, Hold - consume stock
- Available - restores stock
- Cancelled + Refund - restores stock and refunds payment

## 6.9 Coaches Tab

Four sections:

- Assign form - pick coach + pick client, click Assign
- Current coaches table - shows client count, allows demotion
- Active assignments table - shows all pairs, allows removal
- Promote users table - turns regular users into coaches

## 6.10 Inventory Tab

Full CRUD for rental items. Add new equipment, edit rates and stock, or delete items (blocked if active rentals reference them).

## 6.11 Exercises Tab

Full CRUD for the exercise library. Add/edit/delete exercises. Deletion is blocked if the exercise is used in any workout, routine, or PR.

## 7. API Reference

### 7.1 Authentication

| Method | Endpoint           | Description                               |
|--------|--------------------|-------------------------------------------|
| POST   | /api/auth/register | Create account                            |
| POST   | /api/auth/login    | Get JWT token                             |
| GET    | /api/auth/me       | Get current user                          |
| PATCH  | /api/auth/me       | Update profile (name, username, birthday) |

### 7.2 Workouts

| Method | Endpoint          | Description                        |
|--------|-------------------|------------------------------------|
| GET    | /api/workouts     | List user's workouts               |
| GET    | /api/workouts/:id | Get workout with sets              |
| POST   | /api/workouts     | Create workout (with PR detection) |
| DELETE | /api/workouts/:id | Delete workout                     |

### 7.3 Routines

| Method | Endpoint          | Description                |
|--------|-------------------|----------------------------|
| GET    | /api/routines     | List routines              |
| GET    | /api/routines/:id | Get routine with exercises |
| POST   | /api/routines     | Create routine             |
| DELETE | /api/routines/:id | Delete routine             |

### 7.4 Analytics

| Method | Endpoint                      | Description      |
|--------|-------------------------------|------------------|
| GET    | /api/analytics/volume?days=30 | Volume over time |
| GET    | /api/analytics/prs            | Personal records |

|     |                                       |                           |
|-----|---------------------------------------|---------------------------|
| GET | /api/analytics/muscle-balance?days=30 | Muscle group distribution |
|-----|---------------------------------------|---------------------------|

## 7.5 Memberships & Payments

| Method | Endpoint                        | Description                                      |
|--------|---------------------------------|--------------------------------------------------|
| GET    | /api/memberships/me             | User's memberships                               |
| POST   | /api/memberships                | Subscribe                                        |
| POST   | /api/payments                   | Create payment (triggers QR for GCash/Maya/Bank) |
| GET    | /api/payments/me                | User's payment history                           |
| GET    | /api/payments/verify/:reference | Public verification page                         |
| PATCH  | /api/payments/:id/confirm       | User confirms payment sent                       |
| POST   | /api/payments/:id/resend-email  | Resend QR email                                  |

## 7.6 Rentals

| Method | Endpoint                | Description    |
|--------|-------------------------|----------------|
| GET    | /api/rentals/items      | Public catalog |
| POST   | /api/rentals            | Create rental  |
| POST   | /api/rentals/:id/return | Return item    |
| GET    | /api/rentals/me         | User's rentals |

## 7.7 Coaching

| Method | Endpoint                          | Description                     |
|--------|-----------------------------------|---------------------------------|
| GET    | /api/coaches/me                   | Coach profile + stats           |
| GET    | /api/coaches/clients              | List assigned clients           |
| GET    | /api/coaches/clients/:id          | Client detail                   |
| GET    | /api/coaches/clients/:id/workouts | Client workout history          |
| POST   | /api/coaches/feedback             | Send feedback                   |
| GET    | /api/coaches/feedback             | Sent feedback                   |
| GET    | /api/my-coach                     | Client: my coach + unread count |

|       |                                 |                           |
|-------|---------------------------------|---------------------------|
| GET   | /api/my-coach/feedback          | Client: received feedback |
| PATCH | /api/my-coach/feedback/:id/read | Mark as read              |

## 7.8 Admin

| Method | Endpoint                             | Description              |
|--------|--------------------------------------|--------------------------|
| GET    | /api/admin/users                     | All users                |
| PATCH  | /api/admin/users/:id/role            | Change role              |
| GET    | /api/admin/users/:id/details         | Full user drill-down     |
| GET    | /api/admin/stats?range=30            | Platform stats           |
| GET    | /api/admin/revenue?days=30           | Revenue analytics        |
| GET    | /api/admin/activity?limit=100        | Activity log             |
| GET    | /api/admin/memberships               | All memberships          |
| PATCH  | /api/admin/memberships/:id/status    | Change membership status |
| GET    | /api/admin/rentals                   | All rentals              |
| PATCH  | /api/admin/rentals/:id/status        | Change rental status     |
| GET    | /api/admin/rental-items              | List inventory           |
| POST   | /api/admin/rental-items              | Add item                 |
| PATCH  | /api/admin/rental-items/:id          | Edit item                |
| DELETE | /api/admin/rental-items/:id          | Delete item              |
| GET    | /api/admin/exercises                 | List exercises           |
| POST   | /api/admin/exercises                 | Add exercise             |
| PATCH  | /api/admin/exercises/:id             | Edit exercise            |
| DELETE | /api/admin/exercises/:id             | Delete exercise          |
| GET    | /api/admin/payments/pending          | Pending payments         |
| PATCH  | /api/admin/payments/:id/verify       | Verify payment           |
| PATCH  | /api/admin/payments/:id/reject       | Reject payment           |
| POST   | /api/admin/payments/:id/resend-email | Resend QR email          |
| GET    | /api/admin/coaches                   | List coaches             |
| GET    | /api/admin/coach-assignments         | List assignments         |

|        |                                  |                   |
|--------|----------------------------------|-------------------|
| POST   | /api/admin/coach-assignments     | Create assignment |
| DELETE | /api/admin/coach-assignments/:id | Remove assignment |

## 8. Database Schema

### 8.1 MySQL Tables

| Table             | Purpose                                            |
|-------------------|----------------------------------------------------|
| users             | Accounts with role, email, password hash, birthday |
| exercises         | Exercise library (42+ entries)                     |
| routines          | User-created workout templates                     |
| routine_exercises | Exercises within routines (sets, reps, order)      |
| workouts          | Logged training sessions                           |
| workout_sets      | Individual sets (weight x reps)                    |
| personal_records  | Auto-computed 1RMs (unique per user+exercise)      |
| memberships       | Plans with dates and status                        |
| rental_items      | Equipment inventory with stock                     |
| rentals           | Rental transactions with status enum               |
| coach_clients     | Coach <-> client assignments                       |
| coach_feedback    | Coach notes with rating and read tracking          |

### 8.2 MongoDB Collections

| Collection    | Purpose                                              |
|---------------|------------------------------------------------------|
| activity_logs | Every user event (login, workout, payment, feedback) |
| payment_logs  | Payment records with method-specific details         |
| admin_metrics | Daily aggregated counters for dashboard charts       |

### 8.3 Full SQL Schema

Run this in phpMyAdmin to rebuild the database from scratch:

```
DROP DATABASE IF EXISTS fitforge;
CREATE DATABASE fitforge CHARACTER SET utf8mb4;
USE fitforge;

-- See Section 8.1 for full DDL. Abbreviated summary:
CREATE TABLE users (...);
CREATE TABLE exercises (...);
```

```
CREATE TABLE routines (...);
CREATE TABLE routine_exercises (...);
CREATE TABLE workouts (...);
CREATE TABLE workout_sets (...);
CREATE TABLE personal_records (...);
CREATE TABLE memberships (...);
CREATE TABLE rental_items (...);
CREATE TABLE rentals (...);
CREATE TABLE coach_clients (...);
CREATE TABLE coach_feedback (...);

-- (Full schema available in the project's db/migrations folder)
```

## 9. Troubleshooting

### 9.1 Common Startup Issues

#### Backend won't start - 'Cannot find module'

Run `npm install` in the project root. If errors persist, delete `node_modules` and reinstall.

#### Backend crashes with MongoDB error

Check that your IP is whitelisted in MongoDB Atlas -> Network Access. If you're on a home WiFi that changes IPs often, add `0.0.0.0/0` (allow all) for development.

#### MySQL connection refused

Open XAMPP Control Panel and click Start next to MySQL. Confirm port 3306 is not blocked.

### 9.2 Frontend Issues

#### Tailwind styles not applying

Ensure `client/src/index.css` starts with `@import 'tailwindcss'`. For dark mode, verify the `@custom-variant` line is present. Delete `node_modules/.vite` and restart the dev server.

#### Blank white screen

Open browser DevTools (F12) -> Console. Look for red errors. Common culprits: missing import, null reference, or JSX syntax error.

#### API returns 401 Unauthorized

Your JWT has expired or was cleared. Log out and log back in.

#### API returns 500 Internal Server Error

Check the backend terminal - the exact error is printed there. Common causes: missing DB column, foreign key violation, MongoDB connection timeout.

### 9.3 Payment & Email Issues

#### QR email not arriving

- Check Spam / Promotions folder in Gmail
- Verify `EMAIL_USER` and `EMAIL_PASS` in `.env`
- Gmail App Password must be 16 characters, no spaces
- Check backend terminal for '[mailer]' log lines

#### QR code shows 'Payment not found'

The `CLIENT_URL` in `.env` must match where the frontend is running. If testing on a phone, use your PC's LAN IP (e.g., `http://192.168.1.10:5173`).

## 9.4 Database Issues

### 'Table doesn't exist in engine'

Broken InnoDB metadata. Stop MySQL in XAMPP, delete the orphaned .ibd file from C:/xampp/mysql/data/fitforge/, restart MySQL, and recreate the table.

### Foreign key constraint fails on delete

The row is referenced by another table. Delete child rows first, or use ON DELETE CASCADE (already configured for most tables).

## 9.5 Resetting the System

To wipe everything and start fresh:

- In phpMyAdmin: DROP DATABASE fitforge; then re-run the schema
- In MongoDB Atlas: drop the fitforge database
- In the browser: clear localStorage (DevTools -> Application -> Local Storage -> Clear)
- Restart both backend and frontend

# 10. Appendix

## 10.1 Project Folder Structure

```
FitForge/
|-- server.js Express entry point
|-- .env Secrets (not committed)
|-- db/
| |-- mysql.js MySQL pool
| |-- mongo.js MongoDB connection
| +-- migrations/*.sql Schema evolution
|-- middleware/
| +-- auth.js JWT verification
|-- models/
| +-- mongo/ Mongoose schemas
|-- services/ Mailer, activity logger, PR detection
|-- routes/ API endpoints
+-- client/
| |-- public/
| | +-- logo.png Static assets
| +-- src/
| |-- context/ Auth, Theme, Toast providers
| |-- components/ Reusable UI
+-- pages/ Route components
```

## 10.2 Role Permissions Matrix

| Action                  | User | Coach | Admin |
|-------------------------|------|-------|-------|
| Log workouts            | Yes  | Yes   | Yes   |
| View own analytics      | Yes  | Yes   | Yes   |
| Subscribe to membership | Yes  | No    | No    |
| Rent equipment          | Yes  | No    | No    |
| View coach feedback     | Yes  | No    | No    |
| View client workouts    | No   | Yes   | No    |
| Send coaching feedback  | No   | Yes   | No    |
| Manage users            | No   | No    | Yes   |
| Verify payments         | No   | No    | Yes   |
| Manage inventory        | No   | No    | Yes   |
| Assign coaches          | No   | No    | Yes   |

## 10.3 Environment Variables Reference

| Variable       | Purpose                     | Example               |
|----------------|-----------------------------|-----------------------|
| PORT           | Backend HTTP port           | 5000                  |
| MYSQL_HOST     | MySQL host                  | localhost             |
| MYSQL_USER     | MySQL username              | root                  |
| MYSQL_DATABASE | MySQL database name         | fitforge              |
| MONGO_URI      | MongoDB connection string   | mongodb+srv://...     |
| JWT_SECRET     | Secret for signing tokens   | 64-char hex           |
| JWT_EXPIRES_IN | Token lifetime              | 7d                    |
| EMAIL_USER     | Gmail sender address        | you@gmail.com         |
| EMAIL_PASS     | Gmail app password          | 16 chars              |
| EMAIL_CC_ADMIN | Optional CC on emails       | billing@...           |
| CLIENT_URL     | Frontend base URL for links | http://localhost:5173 |

## 10.4 Keyboard Shortcuts

| Shortcut                | Action                      |
|-------------------------|-----------------------------|
| Ctrl+C in terminal      | Stop the running server     |
| Ctrl+Shift+R in browser | Hard refresh (bypass cache) |
| F12 in browser          | Open DevTools               |
| Ctrl+G in VS Code       | Go to line number           |